THE GRUNDY DOMINATION NUMBER IS NOT MULTIPLICATIVE ON STRONG PRODUCTS

DEEP BHATTACHARJEE

Abstract. The strong product of the octahedron with a graph H on 21 vertices of Grundy domination number 10 has Grundy domination number at least 21. This disproves the conjecture of Brešar, Bujtás, Gologranc, Klavžar, Košmrlj, Patkós, Tuza and Vizer that $\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$ for all graphs G and H.

1. Introduction

Let G be a finite simple graph and write $N[v]$ for the closed neighbourhood of a vertex v. A sequence $(v_1, \dots, v_k)$ of vertices of G is *legal* if every v_i dominates a vertex not dominated by its predecessors, that is,

$$N[v_i] \setminus \bigcup_{j<i} N[v_j] \neq \emptyset \qquad (1 \le i \le k);$$

a vertex in this difference is a *footprint* of v_i. The vertices of a legal sequence are distinct. The *Grundy domination number* $\gamma_{\mathrm{gr}}(G)$ is the length of a longest legal sequence; a longest legal sequence always dominates G. The parameter was introduced by Brešar, Gologranc, Milanič, Rall and Rizzi [4], who connected it with the domination game.

In the strong product $G \boxtimes H$ two distinct vertices (g,h) and (g',h') are adjacent when $g' \in N_G[g]$ and $h' \in N_H[h]$; equivalently $N_{G\boxtimes H}[(g,h)] = N_G[g] \times N_H[h]$. Taking a Grundy dominating sequence of G and one of H and listing their pairs in lexicographic order gives a legal sequence of $G \boxtimes H$, so

$$\gamma_{\mathrm{gr}}(G \boxtimes H) \;\ge\; \gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H) \tag{1.1}$$

for all G and H [3, Proposition 21]. In the same paper the authors conjectured that the bound is always attained.

Conjecture 1.1 ([3, Conjecture 22])**.** *For all graphs G and H, $\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H)$.*

The conjecture holds when G is a caterpillar [3] and, more generally, when G is a forest, by the work of Bell, Driscoll, Krop and Wolff [2]. A proof for all graphs was announced by Herrman and Smith [5] and later withdrawn.[1] We show that the conjecture is false.

For $N \ge 1$ let H_N be the graph on $\{0, 1, \dots, N-1\}$ in which u and v are adjacent when $|u - v| \equiv 2 \pmod 3$. Let $K_{2,2,2}$ be the octahedron, the complete tripartite graph with three parts of size two.

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Corresponding author: d.bhattacharjee@erl-forschung.de; ORCID 0000-0003-0466-750X.

[1] The second version of arXiv:2212.04565 (12 January 2023) carries the comment "There is an error. Some cases were not considered in the proof of Theorem 1."

Theorem 1.2. $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$, $\gamma_{\mathrm{gr}}(H_{21}) = 10$ *and* $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_{21}) \geq 21$. *In particular*

$$\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_{21}) \;>\; \gamma_{\mathrm{gr}}(K_{2,2,2})\,\gamma_{\mathrm{gr}}(H_{21}),$$

and Conjecture 1.1 fails.

The lower bound for the product is a short explicit argument valid for every N (Proposition 3.1), and so is $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$ (Lemma 2.1). The upper bound $\gamma_{\mathrm{gr}}(H_{21}) \leq 10$ is the one step that rests on a finite computation: an exhaustive search through the sets dominated by legal sequences, which we have certified in the Lean 4 proof assistant [6] (Section 4). The same search gives a family of counterexamples.

Theorem 1.3. *For every* $N \geq 1$, $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N$. *For* $6 \leq N \leq 63$,

$$\gamma_{\mathrm{gr}}(H_N) = \left\lfloor \frac{N+1}{3} \right\rfloor + 3,$$

so $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) > \gamma_{\mathrm{gr}}(K_{2,2,2})\,\gamma_{\mathrm{gr}}(H_N)$ *for* $21 \leq N \leq 63$. *For* $N = 63$ *the ratio* $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N)/(\gamma_{\mathrm{gr}}(K_{2,2,2})\gamma_{\mathrm{gr}}(H_N))$ *is at least* $63/48 = 21/16$.

The second statement of Theorem 1.3 is computational (Section 5); the first is Proposition 3.1. If the formula for $\gamma_{\mathrm{gr}}(H_N)$ holds for all N, the ratio approaches $3/2$. By Proposition 25 of [3], $\gamma_{\mathrm{gr}}(G \boxtimes H) \leq |V(G)|\,\gamma_{\mathrm{gr}}(H)$, so for $G = K_{2,2,2}$ the ratio can never exceed 3.

2. The graphs

We write the vertices of the octahedron as $g_0, g_1, g_2, x_0, x_1, x_2$ (in this order, $0, \ldots, 5$), where x_t is the antipode of g_t; the three antipodal pairs are the only non-adjacent pairs (Figure 1). Thus $N[v]$ is everything except the antipode of v.

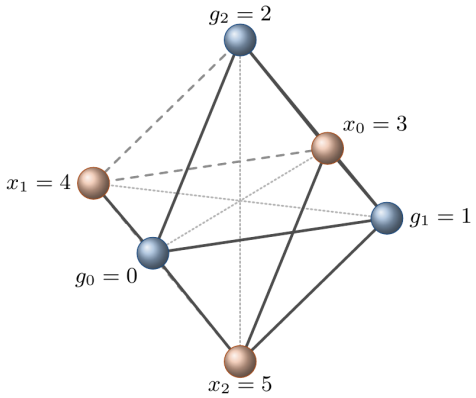


FIGURE 1. The octahedron $K_{2,2,2}$. Each vertex is adjacent to all others except its antipode; the dotted segments join the antipodal pairs $\{g_t, x_t\}$.

Lemma 2.1. $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$.

Proof. The sequence (g_0, x_0) is legal: x_0 footprints itself. Conversely, let (u, v) be legal. If v is the antipode of u, then $N[u] \cup N[v]$ is the whole vertex set; otherwise $N[u]$ misses only the antipode of u, which lies in $N[v]$. Either way nothing is left for a third vertex to footprint. □

In H_N we write $a_i = 3i$, $b_i = 3i+1$ and $c_i = 3i+2$. Two vertices of the same residue class are never adjacent, and for vertices of different classes,

$$a_i \sim c_j \iff j \geq i, \qquad b_i \sim a_j \iff j > i, \qquad c_i \sim b_j \iff j > i, \tag{2.1}$$

because $c_j - a_i = 3(j-i)+2$, $a_j - b_i = 3(j-i)-1$ and $b_j - c_i = 3(j-i)-1$. When $N = 3n$ the graph has a circular description (Figure 2). Recall that the circular clique $K_{p/q}$ has vertex set $\mathbb{Z}_p$, two vertices being adjacent when their cyclic distance is at least q.

Proposition 2.2. *Let $n \geq 2$ and place the vertices of H_{3n} on a cycle in the order*

$$a_0, \ldots, a_{n-1},\ c_0, \ldots, c_{n-1},\ b_0, \ldots, b_{n-1}.$$

Two vertices are adjacent in H_{3n} if and only if their cyclic distance is at least $n+1$, or they are a_i and c_i for some i. Thus H_{3n} is the circular clique $K_{3n/(n+1)}$ together with the perfect matching $\{a_i c_i\}$.

Proof. Give a_i, c_i and b_i the positions i, $n+i$ and $2n+i$ modulo $3n$. Vertices of the same class are at cyclic distance at most $n-1$ and are not adjacent. For two vertices of different classes, name them so that the second follows the first in the cyclic order $A \to C \to B \to A$ of the blocks: the pairs are (a_i, c_j), (c_i, b_j) and (b_i, a_j). In each case the second position minus the first is $n + (j-i)$ modulo $3n$, with $|j-i| \leq n-1$, so the cyclic distance is at least $n+1$ when $j > i$, exactly n when $j = i$, and at most $n-1$ when $j < i$. Comparing with (2.1): the pair is adjacent exactly when $j > i$, except that $a_i \sim c_i$. □

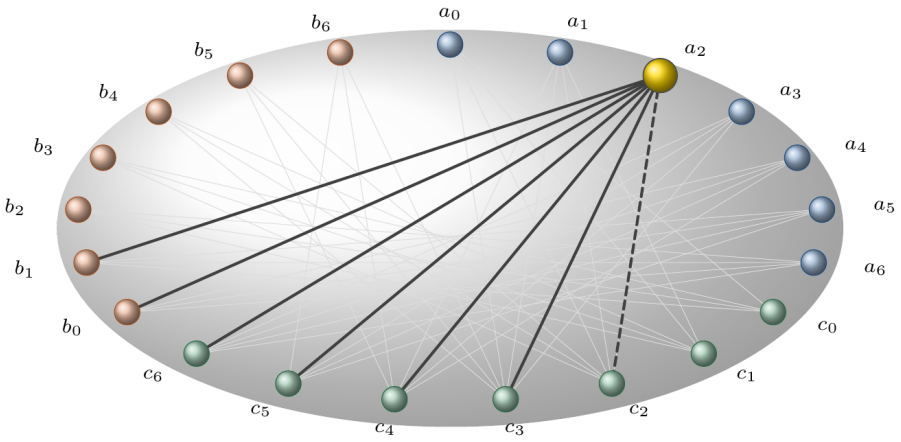


FIGURE 2. H_{21} on a ring in the order of Proposition 2.2; all edges are drawn faintly. The closed neighbourhood of a_2 is a_2 together with the arc $c_3, \ldots, c_6, b_0, b_1$ of vertices at cyclic distance at least 8 and its partner c_2 (dashed).

3. A long legal sequence in the product

The construction uses one property of H_N: with the colouring $v \mapsto v \bmod 3$, every edge uv with $u < v$ satisfies $u \equiv v+1 \pmod 3$. We state it for any graph with this property.

Proposition 3.1. *Let H be a graph with vertices $v_0, \ldots, v_{N-1}$ and a map $t\colon V(H) \to \mathbb{Z}_3$ such that $t(v_i) \equiv t(v_j) + 1 \pmod 3$ for every edge $v_i v_j$ with $i < j$. Then the sequence*

$$s_j = (g_{t(v_j)},\ v_j), \qquad j = 0, 1, \ldots, N-1,$$

is legal in $K_{2,2,2} \boxtimes H$*, and* $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H) \geq N$*. In particular* $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N$ *for every* $N \geq 1$*, with* $t(v) = v \bmod 3$*.*

Proof. We show that s_j footprints $f_j = (x_{t(v_j)+1}, v_j)$, indices of g and x taken modulo 3. First, $f_j \in N[s_j]$: the vertex $x_{t(v_j)+1}$ is not the antipode $x_{t(v_j)}$ of $g_{t(v_j)}$, so it lies in $N[g_{t(v_j)}]$, and $v_j \in N[v_j]$. Now let $i < j$. If $f_j \in N[s_i]$, then $x_{t(v_j)+1} \in N[g_{t(v_i)}]$, which means $t(v_i) \neq t(v_j)+1$, and $v_j \in N[v_i]$, which for $i \neq j$ means that $v_i v_j$ is an edge, hence $t(v_i) = t(v_j) + 1$. The two conclusions contradict each other, so $f_j \notin N[s_i]$. □

Figure 3 shows the sequence for H_{21}. Every column of the product contains exactly one vertex of the sequence, while a Grundy dominating sequence of the form used for (1.1) would use only $\gamma_{\mathrm{gr}}(H_{21}) = 10$ columns. The rotation of the three colours along the sequence is what lets each new vertex reach a footprint in a layer x_t that none of the earlier vertices covers in that column.

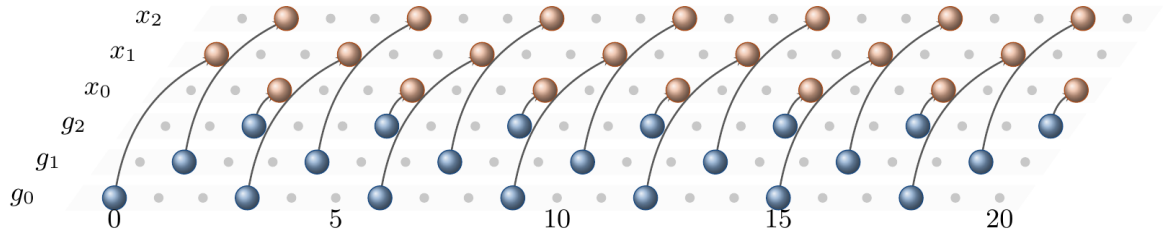


FIGURE 3. $K_{2,2,2} \boxtimes H_{21}$ as six stacked copies of H_{21}, one for each vertex of the octahedron. The 21 blue vertices $(g_{j \bmod 3}, j)$ form a legal sequence; the arrow from each of them ends at its footprint $(x_{(j+1) \bmod 3}, j)$.

4. The Grundy domination number of H_{21}

Lemma 4.1. $\gamma_{\mathrm{gr}}(H_{21}) \geq 10$.

Proof. The sequence $(a_0, a_1, a_2, a_3, a_4, b_6, c_4, c_0, a_5, b_0)$, that is $(0, 3, 6, 9, 12, 19, 14, 2, 15, 1)$, is legal and dominating. By (2.1), a_0 dominates $c_0, \ldots, c_6$; each a_i with $1 \leq i \leq 4$ footprints itself and b_{i-1}; then b_6 footprints itself, c_4 footprints b_5, c_0 footprints b_4, a_5 footprints itself and b_0 footprints a_6. □

For the upper bound we use the sets of dominated vertices. For a legal sequence $S = (v_1, \ldots, v_k)$ put $D(S) = N[v_1] \cup \cdots \cup N[v_k]$. Define families $\mathcal{L}_0, \mathcal{L}_1, \ldots$ of vertex sets by $\mathcal{L}_0 = \{\emptyset\}$ and

$$\mathcal{L}_{k+1} = \{\, D \cup N[v] \;:\; D \in \mathcal{L}_k,\ v \in V(G),\ N[v] \not\subseteq D \,\}.$$

Lemma 4.2. *If* S *is a legal sequence of length* k*, then* $D(S) \in \mathcal{L}_k$*. Hence* $\gamma_{\mathrm{gr}}(G)$ *is the largest* k *with* $\mathcal{L}_k \neq \emptyset$*.*

Proof. Induction on k: if $(v_1, \ldots, v_{k+1})$ is legal, then $D = D(v_1, \ldots, v_k) \in \mathcal{L}_k$ and $N[v_{k+1}] \not\subseteq D$, so $D \cup N[v_{k+1}] \in \mathcal{L}_{k+1}$. Conversely, every member of $\mathcal{L}_k$ is $D(S)$ for a legal sequence S of length k, by the same induction. □

The families can be large in general, but for H_{21} they are small: by (2.1), the vertices of a class dominated by the vertices of another class form an initial or a final segment of that class.[2] Their sizes are

$$|\mathcal{L}_0|, \dots, |\mathcal{L}_{10}| = 1,\ 21,\ 210,\ 1023,\ 1482,\ 1302,\ 848,\ 419,\ 150,\ 22,\ 1,$$

and $\mathcal{L}_{11} = \emptyset$; the only member of $\mathcal{L}_{10}$ is the whole vertex set. With Lemma 4.2 this gives the following.

Proposition 4.3. $\gamma_{\mathrm{gr}}(H_{21}) \le 10$.

We have verified Proposition 4.3 in three independent ways. In Lean 4, a certificate lists the 5479 sets of the families $\mathcal{L}_0, \dots, \mathcal{L}_{10}$ as bit masks stored in binary search trees, and the kernel checks by evaluation that $\emptyset \in \mathcal{L}_0$, that $D \cup N[v] \in \mathcal{L}_{k+1}$ for every $D \in \mathcal{L}_k$ and every v with $N[v] \not\subseteq D$, and that no member of $\mathcal{L}_{10}$ has a proper extension. A formal version of Lemma 4.2, proved in Lean for every graph and every such certificate, then bounds the length of every legal sequence of H_{21} by 10. The Lean statement uses the definitions of legal sequences, of H_{21}, of $K_{2,2,2}$ and of the strong product given in this paper, and the main theorem

> `strong_product_counterexample`: there is a dominating legal sequence of length 21 in $K_{2,2,2} \boxtimes H_{21}$, every legal sequence of $K_{2,2,2}$ has length at most 2, every legal sequence of H_{21} has length at most 10, and $2 \cdot 10 < 21$

depends only on the standard axioms `propext` and `Quot.sound`; no `native_decide` is used. Independently, a C program computes the families by breadth-first search, and a Python program finds $\gamma_{\mathrm{gr}}(H_{21}) = 10$ by a memoised depth-first search over dominated sets; both use only the definition of H_{21}.

Proof of Theorem 1.2. Lemma 2.1, Lemma 4.1 with Proposition 4.3, and Proposition 3.1 with $N = 21$. □

5. The family H_N

The same breadth-first search, run for $6 \le N \le 63$, gives the values in Table 1; for $N \le 24$ they agree with the depth-first search. With Proposition 3.1 this proves Theorem 1.3.[3] The largest search, for $N = 63$, visits 193 131 321 dominated sets and takes about five minutes on one core. The upper bound for $N = 21$ is the only one we have checked in Lean.

Table 1. $\gamma_{\mathrm{gr}}(H_N) = \lfloor (N+1)/3 \rfloor + 3$ for $6 \le N \le 63$, by exhaustive search. The conjecture fails for $K_{2,2,2} \boxtimes H_N$ when $N > 2\gamma_{\mathrm{gr}}(H_N)$, that is, for $N \ge 21$.

N	6–7	8–10	11–13	14–16	17–19	20–22	23–25	26–28	29–31	⋯	59–61	62–63
$\gamma_{\mathrm{gr}}(H_N)$	5	6	7	8	9	10	11	12	13	⋯	23	24

Question 5.1. Is $\gamma_{\mathrm{gr}}(H_N) = \lfloor (N+1)/3 \rfloor + 3$ for every $N \ge 6$? An affirmative answer would give $\sup_{G,H} \gamma_{\mathrm{gr}}(G \boxtimes H)/(\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)) \ge 3/2$.

Question 5.2. Is $\gamma_{\mathrm{gr}}(G \boxtimes H)/(\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H))$ bounded over all graphs G and H? Which graphs G satisfy Conjecture 1.1 for every H? By [2] all forests do, and Theorem 1.2 shows that $K_{2,2,2}$ does not.

[2] For instance, the vertices c_j dominated by the chosen vertices of the class a are those with $j \ge \min\{i : a_i \text{ chosen}\}$.

[3] Grundy domination numbers add over disjoint unions, and $G \boxtimes (H \cup H') = (G \boxtimes H) \cup (G \boxtimes H')$, so disjoint copies of H_{21} already give infinitely many counterexamples, all with ratio 21/20.

Remark 5.3. Deleting the matching $\{a_ic_i\}$ from H_{3n} leaves the circular clique $K_{3n/(n+1)}$ (Proposition 2.2), a circulant graph, and it still satisfies the hypothesis of Proposition 3.1. Our search gives $\gamma_{\mathrm{gr}}(K_{3n/(n+1)}) = n+4$ for $3 \le n \le 11$, one more than $\gamma_{\mathrm{gr}}(H_{3n})$. Hence $K_{2,2,2} \boxtimes K_{27/10}$, $K_{2,2,2} \boxtimes K_{30/11}$ and $K_{2,2,2} \boxtimes K_{33/12}$ are counterexamples as well, with a vertex-transitive second factor; the matching only makes the counterexamples smaller.

Computations and formal verification. The Lean development, the C and Python programs, their outputs and the scripts that rebuild this paper are in the archive [1]. The Lean files (Lean 4.34.1, no external libraries) contain 25 theorems; all of them depend only on the standard axioms.

Use of generative AI. The author used Claude (Anthropic) to check the results by computer and in Lean and to help draft the text; the author verified every step and is responsible for the content.

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Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL),
Bhubaneswar, Odisha 751030, India

Email address: d.bhattacharjee@erl-forschung.de

Email address: itsdeep@live.com

URL: https://orcid.org/0000-0003-0466-750X